

Data Acquisition

Docker on Google Cloud

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CHANGE THE WORLD FROM HERE

Overview

Running Docker Container on Google Cloud

- Create a repo with codes including Dockerfile
- Connect it to Google Cloud Run
- Trigger crontab using Cloud Scheduler





Contents

CloudRun

- Connect GitHub to CloudRun for Continuous Deployment
- Configuration
- Endpoint

Cloud Scheduler

- Define schedule and configure



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GCP Cloud Run

Cloud Run

- A fully managed serverless platform that you can run a container.
 - Serverless : It handles infrastructure and you do not need to manage and scale your servers.
 - Container-based : You can create a container image and deploy it to Cloud Run
 - Auto Scaling : Based on the amount of requests, Cloud Run scales up or down.
 - Cost Effective : You can choose pay-per-use.



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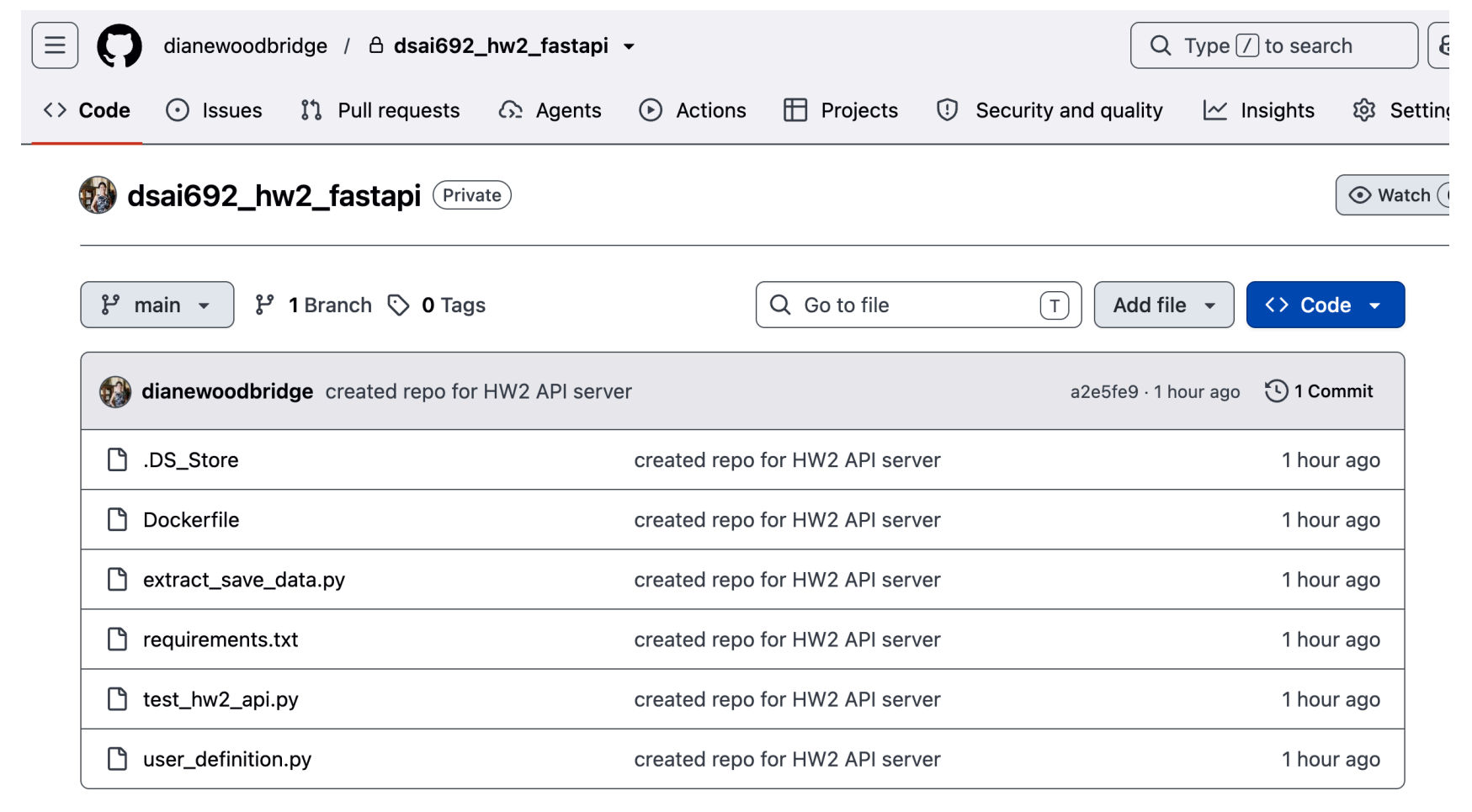
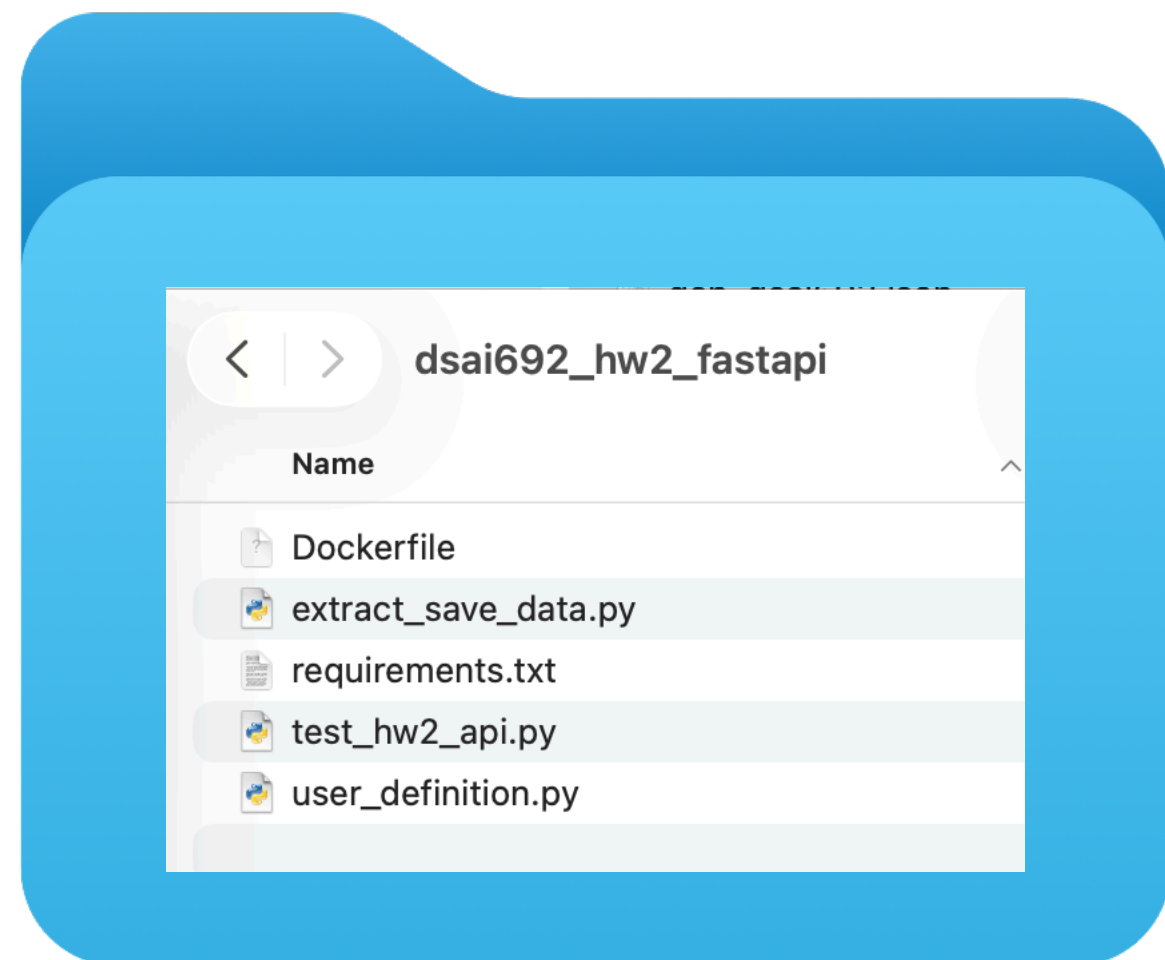
Cloud Scheduler

- Define schedule and configure

Your Microservice to GitHub

Create a repo that contains your code.

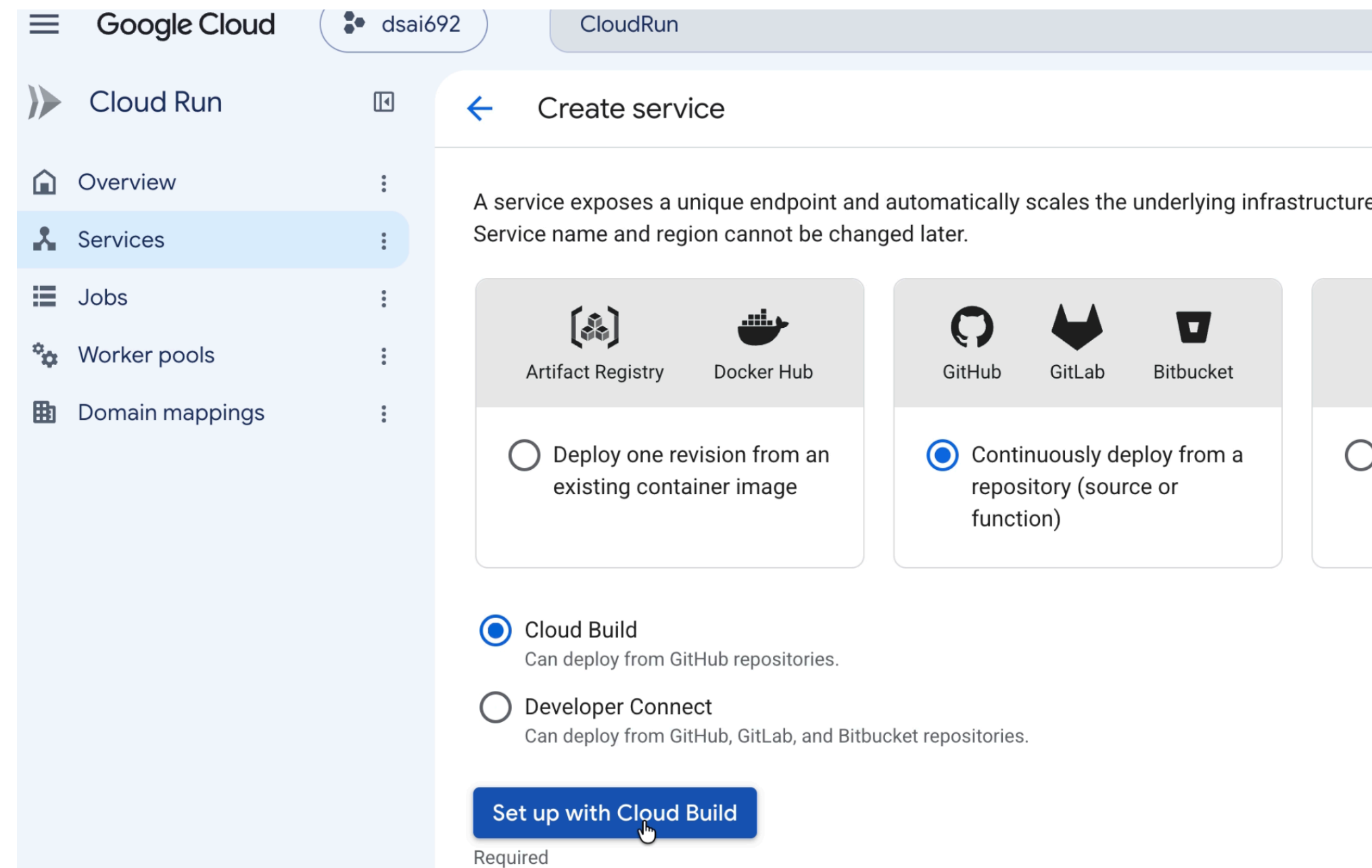
- In this case, I'd place components that you'd want to deploy as a containerized micro service.



Connect GitHub to GCP CloudRun

On CloudRun

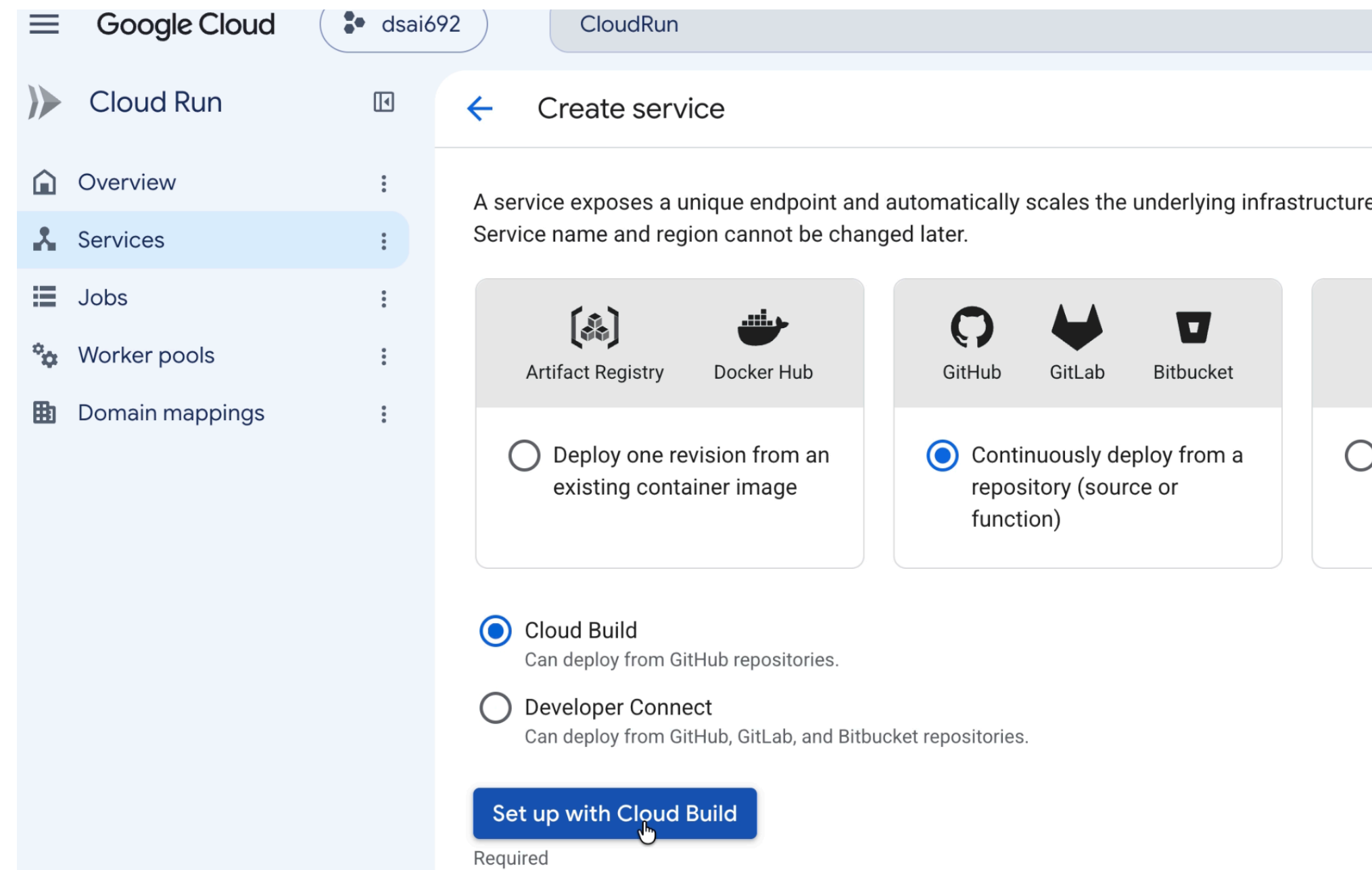
- Go to "Services" tab
- Click "Create a new service"
- Choose "Continuously deploy from a repository"
- Choose "Cloud Build"
- Click "Set up with Cloud Build"
- Authenticate with your Github account and choose the correct repo.



Connect GitHub to GCP CloudRun

On CloudRun

- Go to "Services" tab
- Click "Create a new service"
- Choose "Continuously deploy from a repository"
- Choose "Cloud Build"
- Click "Set up with Cloud Build"
- Authenticate with your Github account and choose the correct repo.



Connect GitHub to GCP CloudRun

On CloudRun

- On "Build Configuration"
 - Choose the branch
 - Build Type as Dockerfile and location.

✓ Source repository

2 Build Configuration

Branch *

Use a regular expression to match to a specific branch [Learn more](#)

Matches the branch: main

Build Type

☒ Dockerfile

Source location *

Location and name of the Dockerfile. The directory will also be used as the Docker build context.

☐ Go, Node.js, Python, Java, .NET Core, Ruby or PHP via [Google Cloud's buildpacks](#)

Save



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Configure on Cloud Run

Choose a region to deploy your microservice

Important

1. Add a port to use
 - Ex. FastAPI - 8000
2. Choose the correct service account key with proper permissions
3. Add environment variables
 - Service key path : should be same as the mounted volume's path
4. Mount the secret as a volume



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^ Edit Container

Source repository

http://github.com/dianewoodbridge/dsai692_hw2_fastapi

Cloud Build trigger

A Cloud Build trigger will be created in order to automatically build and deploy your code.

Container port

8000

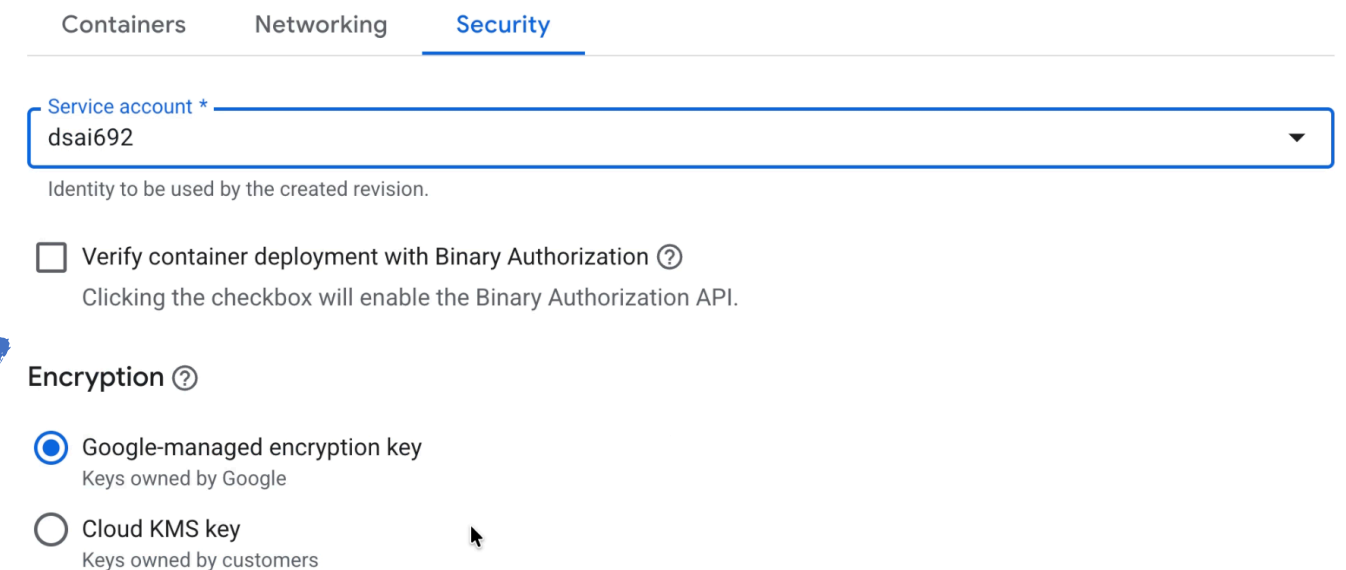
Requests will be sent to the container on this port. We recommend listening on \$PORT instead of this specific number.

Configure on Cloud Run

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Containers Networking **Security**

Service account *
dsai692

Identity to be used by the created revision.

☐ Verify container deployment with Binary Authorization ?
Clicking the checkbox will enable the Binary Authorization API.

Encryption ?

☒ Google-managed encryption key
Keys owned by Google

☐ Cloud KMS key
Keys owned by customers

Configure on Cloud Run

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Settings Variables & Secrets Volumes

Environment variables

Paste a .env file into the name input field to populate environment variables in bulk.

Name 1 GCP_PROJECT_ID e.g. ENV, or paste .env file	Value 1 dsai692 e.g. prod
Name 2 VERTEX_AI_PROJECT_ID e.g. ENV, or paste .env file	Value 2 555299671262 e.g. prod
Name 3 SEARCH_ENGINE_ID e.g. ENV, or paste .env file	Value 3 dsai-hw_1787542785281 e.g. prod

+ Add variable

Configure on Cloud Run

Choose a region to deploy your microservice

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4. Mount the secret as a volume

Create Secret

properties, visit [Secret Manager](#).

If you specify a secret value, it will be in the first version. [Learn more](#)

Name *

dsai692-gcp-key

The name should be identifiable and unique within this project.

Secret value

Input your secret value or import it directly from a file.

Upload file

gcp_dsai692.json



Browse

Maximum size: 64 KiB

Secret value

```
{  
  "type": "service_account",  
  "project_id": "dsai692",  
}
```



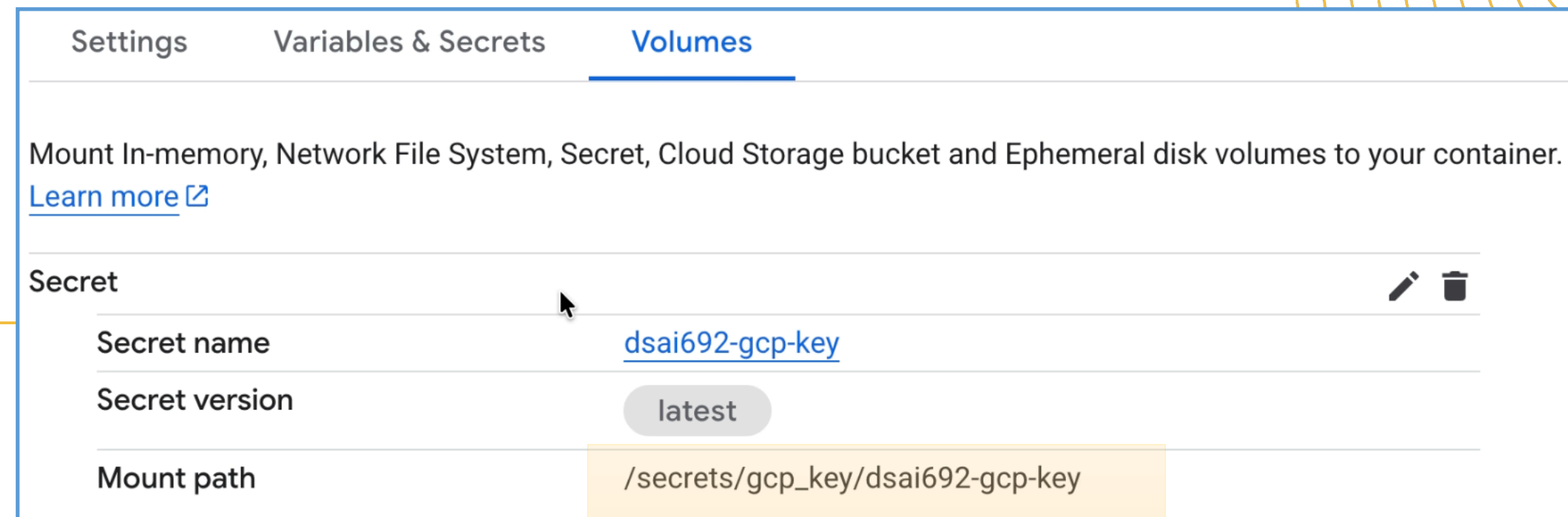
CRC-32C checksum: 0x167AF3EC

Configure on

Choose a region to deploy your microservice

Important

1. Add a port to use
 - Ex. FastAPI - 8000
2. Choose the correct service account key with proper permissions
3. Add environment variables
 - Service key path : should be same as the mounted volume's path
4. Mount the secret as a volume

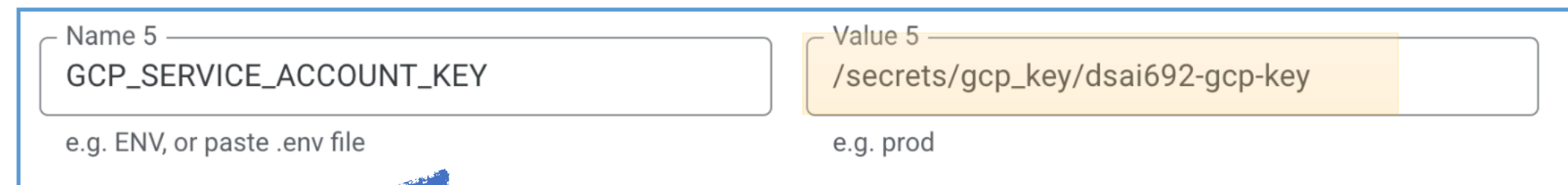


Settings Variables & Secrets **Volumes**

Mount In-memory, Network File System, Secret, Cloud Storage bucket and Ephemeral disk volumes to your container. [Learn more](#)

Secret

Secret name	dsai692-gcp-key
Secret version	latest
Mount path	/secrets/gcp_key/dsai692-gcp-key



Name 5 — GCP_SERVICE_ACCOUNT_KEY

Value 5 — /secrets/gcp_key/dsai692-gcp-key

e.g. ENV, or paste .env file e.g. prod

Configure on Cloud Run

Cloud Run

Services

Jobs

Worker pools

Domain mappings

Create service

Add Container

Requests

Request timeout

300

seconds

Time within which a response must be returned (maximum 3600 seconds).

Maximum concurrent requests per instance

80

The maximum number of concurrent requests that can reach each instance. [What is concurrency?](#)

Execution environment

The execution environment your container runs in. [Learn More](#)

☒ Default

Cloud Run will select a suitable execution environment for you.

☐ First generation

Faster cold starts.

☐ Second generation

Network file system support, full Linux compatibility, faster CPU and network performance.

Revision scaling

Minimum and maximum numbers of instances for the new revision.

Minimum number of instances *

0

Maximum number of instances *

1

The service minimum instances is preferable for most use-cases. Only use this setting if you specifically require per-revision settings.

☒ Startup CPU boost

Start containers faster by allocating more CPU during startup time. [Learn more](#)

Cloud SQL connections

Pricing summary

Cloud Run pricing

Free tier

First 180,000 vCPU-seconds/month

First 360,000 GiB-seconds/month

2 million requests/month

[Check paid tiers details](#)

[Open pricing calculator](#)

I set maximum number of instances as 1, to avoid unexpected high \$\$.



Do this using gcloud

1. Install gcloud CLI to your computer

- <https://docs.cloud.google.com/sdk/docs/install-sdk>

2. Initialize gcloud CLI

- `gcloud init`

3. Write gcloud code

```
API_SERVICE=hw2-api-server
LOCAL_KEY_FILE=/Users/dwoodbridge/.ssh/gcp_dsai692.json
ENV_FILE=/Users/dwoodbridge/Class/2026_MSDS692/Individual_HW/.env
GCP_KEY_SECRET=hw2-gcp-key

if ! gcloud secrets describe "$GCP_KEY_SECRET" >/dev/null 2>&1; then
  gcloud secrets create "$GCP_KEY_SECRET" --replication-policy=automatic
fi
gcloud secrets versions add "$GCP_KEY_SECRET" --data-file="$LOCAL_KEY_FILE"

gcloud run deploy "$API_SERVICE" \
  --source ./fastapi \
  --region us-west1 \
  --port=8000 \
  --allow-unauthenticated \
  --env-vars-file="$ENV_FILE" \
  --set-secrets="/run/secrets/gcp_key=$GCP_KEY_SECRET:latest"
```



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Cloud Scheduler

- Define schedule and configure

Docker Container to Cloud Run

Google Cloud

Data Acquisition Project

Search (/) for resources, docs, products, and more

Search

✦ 🗨 6 ? ⋮ 👤

Cloud Run

Services

Jobs

Worker pools

Domain mappings

Service details

Edit & deploy new revision

Connect to repo

Test

Learn

Refresh

Deploying revision

Hide status

Updating service

Completed

Creating revision

Completed

Routing traffic

Completed

msds501

Region: us-west1

URL: <https://msds501-477009951698.us-west1.run.app>

Auto (Min: 0, Max: 1)

Metrics

SLOs

Logs

Revisions

Triggers

Networking

Security

YAML

Revisions

Manage traffic

Filter

Filter revisions

Name	Traffic	Deployed	Revision tags	Actions
msds501-00005-57q	100% (to latest)	Just now	+	⋮

msds501-00005-57q

Deployed by dianemkwoodbridge@gmail.com using Cloud Console

Containers

Volumes

Networking

Security

YAML

General

Billing	Request-based
Startup CPU boost	Enabled
Concurrency	80
Request timeout	300 seconds
Execution environment	Default

Autoscaling

Revision max instances	1
------------------------	---

Image	docker.io/dwoodbridge/msds501@sha256:f429e226084fe93c2e193...
Port	8501
Build	(no build information available)
Source	(no source information available)
Command and args	(container entrypoint)
CPU limit	1
Memory limit	128MiB

Use the URL once everything is completed.



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Cloud Scheduler

Using GCP Cloud Scheduler, you can schedule a task to be executed at defined times or regular intervals



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Cloud Scheduler

- **Define schedule and configure**

Trigger crontab using Cloud Scheduler

Go to Cloud Scheduler => Create Job

- Define Schedule and name
- Configure appropriately
 - Set methods and body
 - Make sure to have headers be
 - Name: Content-Type, Value:application/json

Google Cloud Data Acquisition Project Search (/)

Cloud Scheduler / Jobs / Create job

Create a job

 Schedule more targets with Workflows **New**
Combine Google Cloud services and APIs to easily build reliable applications.
[Learn more](#)

☒ Define the schedule

Name *
crontab-example
Must be unique across the jobs in the same region

Region *
us-central1 (Iowa)

Description

Frequency *

Schedules are specified using unix-cron format. E.g. every minute: "* * * * *", every 3 hours: "0 */3 * * *", every Monday at 9:00: "0 9 * * 1". [Learn more](#)

i Minute:
◦ Every minute

Timezone *
Pacific Daylight Time (PDT)

i Jobs in set in timezones affected by Daylight Saving Time can run outside of cadence during DST change. Using a UTC timezone can avoid the problem. [Learn more](#)

[Continue](#)

Trigger cron tab using Cloud Scheduler

Go to Cloud Scheduler => Create Job

- Define Schedule and name
- Configure appropriately
 - Set methods and body
 - Make sure to have headers be
 - Name: Content-Type, Value: application/json

Configure the execution

Target type *

HTTP

URL *

https://dwoodbridg-hw3-fastapi-477009951698.europe-west1.run.app/s

HTTP method

POST

HTTP headers

Some headers are set to default values or removed by Cloud Scheduler.

[Learn more](#)

Name 1 *

Content-Type

Value 1

application/json

+ Add a header

Body

icrosoft.com/", "Google": "https://www.google.com/about/careers/appl
ications/jobs"}}

Auth header

None

Continue

Trigger crontab using Cloud Scheduler

Make sure to run it (force to run if the interval is too big) to ensure it was set correctly.

Jobs

Create job

Refresh

Force run

Edit

Copy

Pause

Resume

Delete

Learn

Scheduler jobs

App Engine Cron jobs

Filter

Filter jobs

?

⋮

☐	Name ↑	Status of last execution	Region	State	Description	Frequency	Target	Last run	Next run	Last updated	Actions
☐	crontab-example	⋯ Has not run yet	us-central1	Enabled		***** (America/Los_Angeles)	URL : https://dwoodbridg-hw3-fastapi-477009951698.europe-west1.run.app/search_and_save/jobs		Sep 14, 2025, 7:24:00 PM	Sep 14, 2025, 7:23:42 PM	⋮ ▶ Force run ≡ View logs
☐	hw3_crontab	✔ Success	us-central1	Enabled		0 0 *** (America/Los_Angeles)	URL : https://dwoodbridg-hw3-fastapi-477009951698.europe-west1.run.app/search_and_save/jobs	Sep 14, 2025, 3:53:49 PM	Sep 15, 2025, 12:00:00 AM	Sep 14, 2025, 3:53:45 PM	

Do this using gcloud

1. Write gcloud code

```
API_SERVICE=hw2-api-server
SCHEDULER_JOB=hw2-refresh-jobs-trigger

if gcloud scheduler jobs describe "$SCHEDULER_JOB" \
    --location=us-west1 >/dev/null 2>&1; then
    SCHEDULER_ACTION=update
else
    SCHEDULER_ACTION=create
fi

gcloud scheduler jobs "$SCHEDULER_ACTION" http "$SCHEDULER_JOB" \
    --location=us-west1 \
    --schedule="0 0 * * *" \
    --uri="$API_SERVER_URL/search_and_save/jobs" \
    --http-method=POST \
    --headers="Content-Type=application/json" \
    --message-body='{"job_title":"Data Engineer", "company_dict":{
    "Google":"www.google.com/about/careers/applications/jobs",
    "OpenAI":"openai.com/careers",
    "Anthropic":"anthropic.com/careers/jobs"}}'
```



References

Docker Hub : <https://docs.docker.com/docker-hub/>

Cloud Run : <https://cloud.google.com/run/docs>

Cloud Scheduler : <https://cloud.google.com/scheduler/docs>